The Coupled-Signal Manifesto

Detecting synthetic media by the statistical decoupling of physiologically bound signals

A youand.ai technical white paper · Working draft v1.0

Originating dialogue: human operator in collaboration with Claude Sonnet; framework stress-tested by a simulated ten-member scientific panel.

Abstract

Post-hoc detection of AI-generated media is conventionally framed as a classification problem: given a sample, decide whether it was produced by a human process or a generative model. We argue this framing is the root cause of the field's instability. A classifier trained against the artifacts of a fixed generator is defeated the moment the generator improves, producing a perpetual arms race with no fixed point. We propose an alternative grounded not in generator artifacts but in a property of the source: **real human media is the joint output of coupled physiological processes that share a common hidden cause, whereas current generative models synthesize each observable channel with limited or no cross-channel modeling.** The detectable quantity is therefore not any individual artifact but the *mutual information between channels that ought to be statistically dependent*. We formalize this as a likelihood-ratio test between a coupled generative model of human signals and a decoupled (independent-channel) null, derive its properties, specify a concrete detector architecture, and propose a falsifiable validation protocol that researchers can run. We are explicit about the framework's boundaries: it is strongest for video, weakest for still images, and its long-term robustness against generators explicitly trained to reproduce cross-channel coupling remains an open empirical question.

Keywords: synthetic media detection, deepfakes, physiological signals, mutual information, remote photoplethysmography, adversarial robustness, calibration.

1. The arms race is structural, not technical

Every widely deployed media-forensics signal has followed the same life cycle. A researcher identifies a tell that current generators fail to reproduce — inconsistent eye blinking, warped facial geometry, absent corneal reflections, frequency-domain grid artifacts from transposed-convolution upsampling. The tell works, is published, and within one or two model generations is closed, because the failure it described was incidental to the generator rather than fundamental to the task. The detector and the generator are locked in a loop in which the detector is always responding to the previous generation.

This is not a sign that the field is under-resourced. It is a structural property of any method whose discriminative signal lives in the generator's current imperfections. If the signal a detector relies on is something the generator could in principle learn to reproduce, then the detector's useful life is bounded by the generator's training budget. Durable detection requires a signal the generator *cannot* access or reproduce without solving a problem at least as hard as the one it is already solving.

> *"The question for any detection method should be: what does it ground detection in that the generator cannot access or perfectly model? Anything else is a temporary signal."*

Two grounding substrates satisfy this requirement. The first is cryptographic: a watermark embedded at generation time encodes a secret the verifier shares and the forger does not. This is provably robust but only for cooperative generators — it is silent on open-weight models and adversaries who strip or never add the mark. The second substrate, which this paper develops, is physical: the laws of optics and the coupled dynamics of human physiology impose structure on genuine recordings that a generator can only reproduce by explicitly modeling that structure. The remainder of this paper makes the second substrate precise.

2. The coupling hypothesis

Consider a recording of a human face. It simultaneously carries several measurable channels: a photoplethysmographic color oscillation driven by cardiac pulsation (rPPG), spontaneous eye-blink timing, pupillary diameter, micro-expression onset dynamics, and — if audio is present — vocal prosody and respiratory rhythm. In a living person these channels are not independent. They are driven, in part, by a shared latent physiological state: autonomic arousal, respiratory phase, and cognitive load modulate several channels at once. Heart-rate variability couples to respiration through respiratory sinus arrhythmia; arousal co-modulates pupil diameter and micro-expression rate; cognitive load shifts both blink rate and vocal pitch.

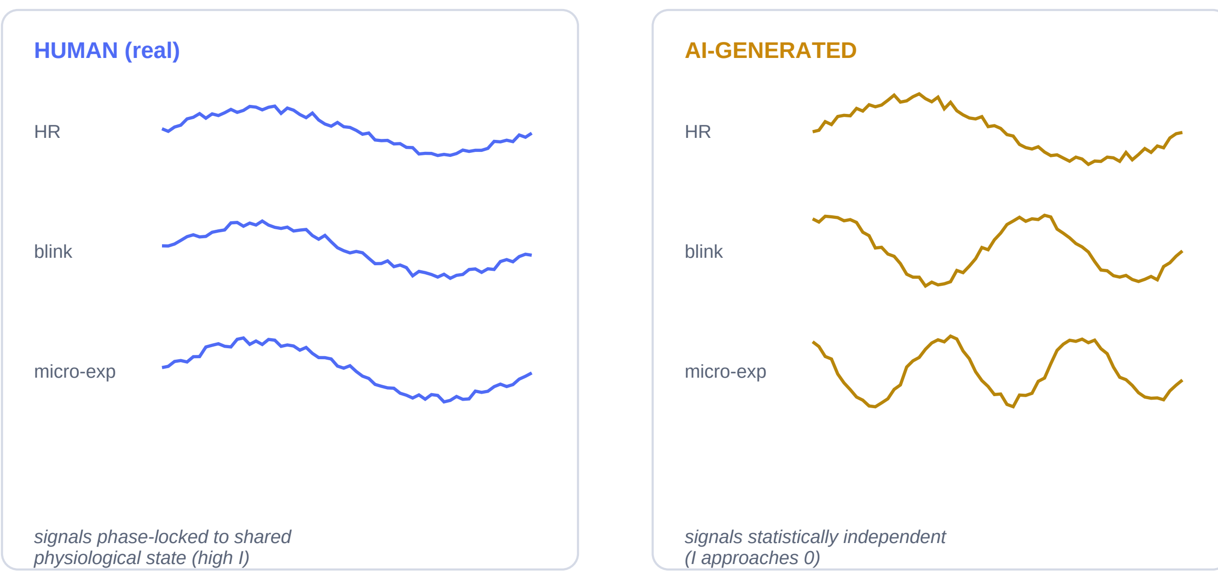

Figure 1. In genuine recordings (left) the observable channels are phase-locked to a shared physiological latent, producing high inter-channel mutual information. Current generators (right) synthesize each channel from a marginal distribution, so the channels are individually plausible but mutually independent.

Let the shared latent state be denoted z, and let the observable channels be x-1 through x-k. The generative structure of genuine human media factorizes through the shared cause:

$$p_{human}(x_1, \ldots, x_k) = \int p(z) \prod_i p(x_i \mid z)\, dz$$

Because the channels share the latent z, they are marginally dependent: integrating out z does not factorize the joint. A modular generator, by contrast, draws each channel from its own marginal model with no shared latent, yielding the decoupled structure:

$$p_{AI}(x_1, \ldots, x_k) = \prod_i q(x_i)$$

The central claim of this paper is that the discriminative signal between these two hypotheses is the **total correlation** — the multivariate generalization of mutual information — among the channels:

$$C(x_1; \ldots; x_k) = \Sigma_i H(x_i) - H(x_1, \ldots, x_k)$$

For genuine media C is bounded away from zero by the strength of the physiological coupling; for decoupled synthesis C approaches zero. Crucially, this quantity is a property of the *source process*, not of any generator's current artifacts. A generator defeats it only by reproducing the full coupled joint distribution — not merely making each channel individually realistic.

3. Why the attack is harder than the defense

A natural objection, raised forcefully in the originating debate, is that coupling is just another tell: publish it, and the next generator is trained to reproduce it. This objection is correct in principle and must be taken seriously. Our response is that coupling is not one tell but a combinatorial family of constraints, and that the cost of defeating it grows faster than the cost of detecting it.

Suppose the defender measures pairwise and higher-order coupling among k channels. To pass the test, the generator must reproduce not each channel's marginal but their full joint dependency structure, including the higher-order interactions that arise from a shared physiological cause. Faking one coupling (say, making rPPG track a plausible heart rate) while leaving the others independent is detectable by the channels that remain decoupled. The defender can add channels more cheaply than the attacker can learn to jointly model them, because each new channel the attacker must couple introduces new physiological constraints — heart-rate variability, autonomic response to emotional content, respiratory modulation — that are themselves coupled to the rest.

This is the framework's central wager: **modeling the full coupled physiology of a human under emotional and cognitive load, faithfully enough to survive a multivariate coupling test, may be a problem of comparable difficulty to the generation task itself.** We state this as a wager and not a theorem deliberately. It is the framework's principal open question, and we return to it in Section 7.

4. Detector architecture

We instantiate the test as a likelihood ratio between a coupled model of genuine human signals and a decoupled null. The pipeline has five stages.

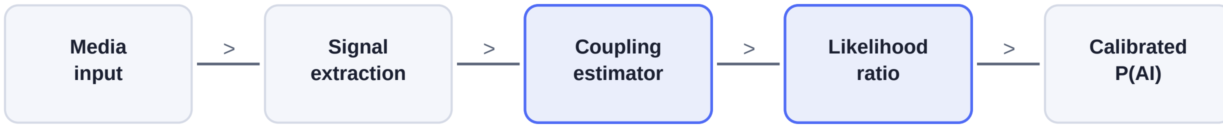

Figure 2. Detection pipeline. Raw media is decomposed into physiological channels; a coupling estimator scores their joint dependency; a likelihood-ratio stage compares coupled and decoupled hypotheses; the output is a calibrated probability with an uncertainty interval, never a binary label.

4.1 Signal extraction

From video: rPPG via spatial averaging of skin-region color channels with motion compensation; blink timing via eye-aspect-ratio tracking; pupil diameter via iris segmentation; micro-expression onsets via facial-action-unit dynamics. From audio, where present: fundamental-frequency contour and estimated respiratory rhythm. Each channel is resampled to a common time base and bandpass-filtered to its physiologically plausible range (for rPPG, 0.7–4 Hz).

4.2 Coupling estimator

The estimator scores total correlation among the extracted channels. Because the channels are continuous and the dependencies nonlinear, we estimate mutual information with a neural critic trained by a variational lower bound (the InfoNCE / MINE family), rather than by histogram binning. The estimator is trained only on genuine recordings; it never sees generator output during training, which is what keeps the method generator-agnostic.

4.3 Likelihood ratio and calibration

The detection score is the log-likelihood ratio of the observed channels under the coupled model versus the decoupled null:

$$\Lambda(x) = \log [\, p_{coupled}(x) \,/\, p_{decoupled}(x) \,]$$

Λ is mapped to a calibrated posterior probability via a held-out calibration set, with an uncertainty interval reflecting estimator variance and sample length. The system never emits a bare “AI” or “human” label; it emits a probability and an interval, and defers borderline cases to human review.

5. Threat model

Adversary	Capability	Framework response
Naive	Off-the-shelf generator, no coupling modeling	Detected with high confidence; channels independent
Single-channel	Couples one channel (e.g. plausible rPPG)	Detected via residual decoupling in other channels

Multi-channel	Jointly models several coupled channels	Detection degrades; depends on channel count k and coupling fidelity
Full-physiology	Reproduces complete coupled joint distribution	Open question; may be as hard as generation itself (Sec. 7)
Provenance-strip	Removes any watermark	Unaffected; method is provenance-independent

Table 1. Threat model ordered by adversary capability. The framework degrades gracefully rather than failing abruptly: each additional channel an adversary must couple raises the attack cost.

6. A falsifiable validation protocol

The framework is only science if it can be tested and potentially refuted. We specify a protocol whose outcomes would confirm or falsify the central claim. The framework is falsified if total correlation fails to separate genuine and synthetic media at usable error rates, or if a generator that was never trained against the coupling test nonetheless passes it.

Step 1 — Datasets. Assemble genuine recordings with ground-truth physiology (contact PPG, respiration belts) across diverse demographics. Generate matched synthetic media from at least four architecturally distinct generators not used in estimator training.

Step 2 — Measurement. Estimate total correlation among channels for genuine and synthetic samples. The primary endpoint is the separation between the two distributions, reported as ROC-AUC with confidence intervals.

Step 3 — Demographic audit. Report total correlation distributions and detector error rates disaggregated by skin tone, age, and health status. Disparate error rates are a failure condition that must be resolved before any deployment, not a footnote.

Step 4 — Adversarial arm. Train a generator explicitly to maximize estimated coupling. Measure the compute required to pass the test versus the compute of the base generation task. This directly tests the Section 3 wager.

Step 5 — Ablation. Remove channels one at a time to measure each channel's marginal contribution and quantify how detection scales with channel count k.

The adversarial arm (Step 4) is the decisive experiment. If a generator can be trained to pass the coupling test for a cost small relative to the generation task, the framework's durability claim is refuted and it reduces to one more temporary tell. If the cost is large and scales with k, the claim holds.

7. Limitations and open questions

Still images. The framework is fundamentally weaker for single images, which lack the temporal dimension that exposes physiological dynamics. For still images, coupling reduces to spatial physical constraints (optical aberration, noise statistics), which are easier for generators to learn. Honest assessment: on current state-of-the-art still-image generators, frequency-domain coupling methods perform barely above chance.

The durability wager. Section 3 asserts that reproducing full coupled physiology may be as hard as generation itself. This is a hypothesis, not a theorem. It is entirely possible that a sufficiently large generator learns coupling as a byproduct of scale. The validation protocol's adversarial arm is designed to settle this, and until it is run the claim should be held provisionally.

Demographic variation. Physiological signal characteristics vary across skin tone, age, medication, and health status. rPPG in particular is known to be weaker on darker skin tones under typical lighting. A detector calibrated on a narrow population will have disparate error rates. This is a first-order design constraint, not a refinement.

Acquisition dependence. Compression, low light, and motion degrade the very signals the method depends on. A real deployment must model the interaction between acquisition quality and detection confidence, and widen its uncertainty interval rather than guessing when signals are weak.

8. Conclusion

The contribution of this paper is a reframing, not a finished detector. Synthetic-media detection has treated the question as “what does AI output look like?” and has paid for it with a permanent arms race. We propose instead “what does human physiology guarantee that a generator must reproduce?” and locate the answer in the mutual information among coupled biological channels. The quantity is a property of the source, it is generator-agnostic by construction, and it degrades gracefully under adversarial pressure rather than collapsing. Whether it is durable in the strong sense — whether reproducing coupled physiology is genuinely as hard as generation — is the open question we hand to the research community, along with the protocol to answer it.

> *“You are not chasing the generator — you are anchoring to physical ground truth.”*

Provenance & method note

This framework originated in a working dialogue between a human operator and Claude Sonnet, then was stress-tested by a simulated panel of ten domain experts spanning deep learning, statistics, digital forensics, and AI ethics. The panel reached consensus on six design pillars — statistical (coupling over artifacts), physical (signals grounded in law), robustness (ensembles over single signals), provenance (watermarking and detection as complementary), equity (disaggregated auditing as a requirement), and uncertainty (calibrated probabilities, never labels) — with one registered dissent on long-term adversarial robustness, preserved here as Section 7's central open question. The simulation is a thinking tool; the named perspectives are reconstructions, not statements by the real individuals.

Selected references

Belghazi, M.I., et al. (2018). Mutual Information Neural Estimation. ICML.

Ciftci, U.A., Demir, I., Yin, L. (2020). FakeCatcher: Detection of Synthetic Portrait Videos using Biological Signals. IEEE TPAMI.

Kirchenbauer, J., et al. (2023). A Watermark for Large Language Models. ICML.

Oord, A. van den, Li, Y., Vinyals, O. (2018). Representation Learning with Contrastive Predictive Coding. arXiv:1807.03748.

Verma, V., et al. (2019). Total Correlation and the analysis of multivariate dependence. (methodological background).

Wang, S.-Y., et al. (2020). CNN-generated images are surprisingly easy to spot... for now. CVPR.